Assignment 3

August 30, 2026

Given:
Collision happening at the starting point (origin: $t = 0, x = 0$). It creates two unstable particles: a neutron and an antimuon (μ^+).
* Frame S is us the observers
* Frame S' is the neutron's rest frame where it isn't moving
* Frame S'' is the antimuon's rest frame
* S' moves at velocity v_n relative to S

1 Part 1

For the neutron, its proper frame is S'. Since the neutron is at rest in its own proper frame, the decay event A occurs at its origin.

$$x'_A = 0$$

For the antimuon, its proper frame is S''. the antimuon is at rest in this frame meaning the decay event B occurs at its origin.

$$x''_B = 0$$

2 Part 2

In S', the time elapsed between creation ($t' = 0$) and decay event A is its proper lifetime:

$$t'_A = \tau_n$$

3 Part3

In S'', the time elapsed until decay event B is its proper lifetime:

$$t''_B = \tau_\mu$$

4 Part4

Using the inverse Lorentz transformation from S' to S:

$$t_A = \gamma_n(t'_A + \frac{v_n x'_A}{c^2})$$

Since $x'_A = 0$ and $t'_A = \tau_n$:

$$t_A = \gamma_n \tau_n$$

5 Part 5

Using the inverse Lorentz transformation from S' to S:

$$x_A = \gamma_n(x'_A + v_n t'_A)$$

Since $x'_A = 0$:

$$x_A = \gamma_n v_n \tau_n$$

6 part 6

Using the inverse Lorentz transformation from S'' to S:

$$t_B = \gamma_\mu \left(t''_B + \frac{v_\mu x''_B}{c^2} \right)$$

Since $t''_B = \tau_\mu$ and $x''_B = 0$:

$$t_B = \gamma_\mu \tau_\mu$$

7 Part 7

Using the inverse Lorentz transformations from S'' to S:

$$x_B = \gamma_\mu(x''_B + v_\mu t''_B)$$

since $x''_B = 0$ and $t_B = \tau_\mu$,

$$x_B = \gamma_\mu(0 + v_\mu \tau_\mu) = \gamma_\mu v_\mu \tau_\mu$$

8 Part 8

Using standard Lorentz transformations from S to S'':

$$t''_A = \gamma_\mu \left(t_A - \frac{v_\mu x_A}{c^2} \right)$$

substituting $t_A = \gamma_n \tau_n$ and $x_A = \gamma_n v_n \tau_n$,

$$t''_A = \gamma_\mu \left(\gamma_n \tau_n - \frac{v_\mu(\gamma_n v_n \tau_n)}{c^2} \right)$$

$$t''_A = \gamma_\mu \gamma_n \tau_n \left(1 - \frac{v_\mu v_n}{c^2} \right)$$

9 Part 9

Using standard Lorentz transformations from S to S'':

$$x''_A = \gamma_\mu(x_A - v_\mu t_A)$$

substituting $x_A = \gamma_n v_n \tau_n$ and $t_A = \gamma_n \tau_n$,

$$x''_A = \gamma_\mu(\gamma_n v_n \tau_n - v_\mu \gamma_n \tau_n)$$

$$x''_A = \gamma_\mu \gamma_n \tau_n (v_n - v_\mu)$$

10 Part 10

Using Lorentz transformations from S to S':

$$t'_B = \gamma_n \left(t_B - \frac{v_n x_B}{c^2} \right)$$

substituting $t_B = \gamma_\mu \tau_\mu$ and $x_B = \gamma_\mu v_\mu \tau_\mu$,

$$t'_B = \gamma_n \left(\gamma_\mu \tau_\mu - \frac{v_n(\gamma_\mu v_\mu \tau_\mu)}{c^2} \right)$$

$$t'_B = \gamma_n \gamma_\mu \tau_\mu \left(1 - \frac{v_n v_\mu}{c^2} \right)$$

11 Part 11

Using Lorentz transformations from S to S':

$$x'_B = \gamma_n(x_B - v_n t_B)$$

substituting $x_B = \gamma_\mu v_\mu \tau_\mu$ and $t_B = \gamma_\mu \tau_\mu$,

$$x'_B = \gamma_n(\gamma_\mu v_\mu \tau_\mu - v_n \gamma_\mu \tau_\mu)$$

$$x'_B = \gamma_n \gamma_\mu \tau_\mu (v_\mu - v_n)$$

12 Part 12

For the neutron to decay before the antimuon as observed in S, we require:

$$t_A < t_B$$

substituting $t_A = \gamma_n \tau_n$ and $t_B = \gamma_\mu \tau_\mu$,

$$\gamma_n \tau_n < \gamma_\mu \tau_\mu$$

expanding $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$,

$$\frac{\tau_n}{\sqrt{1-\frac{v_n^2}{c^2}}} < \frac{\tau_\mu}{\sqrt{1-\frac{v_\mu^2}{c^2}}}$$

assuming the neutron is at rest in S' and setting $v_n = 0$, squaring both sides, and defining the threshold condition for v_μ^{min}:

$$\tau_n = \frac{\tau_\mu}{\sqrt{1-\frac{(v_\mu^{min})^2}{c^2}}}$$

rearranging:

$$1 - \frac{(v_\mu^{min})^2}{c^2} = \frac{\tau_\mu^2}{\tau_n^2}$$

$$\frac{v_\mu^{min}}{c} = \sqrt{1-\frac{\tau_\mu^2}{\tau_n^2}}$$

since $\tau_n \gg \tau_\mu$, $\frac{\tau_\mu^2}{\tau_n^2}$ is much less than 1, applying the binomial approximation ($\sqrt{1-x} \approx 1 - \frac{1}{2}x$ for $x \ll 1$): substitute $x = \frac{\tau_\mu^2}{\tau_n^2}$,

$$\frac{v_\mu^{min}}{c} \approx 1 - \frac{1}{2}\left(\frac{\tau_\mu^2}{\tau_n^2}\right)$$

$$1 - \frac{v_\mu^{min}}{c} \approx \frac{\tau_\mu^2}{2\tau_n^2}$$

given $\tau_n \simeq 879.4$ s and $\tau_\mu \simeq 2.2 \times 10^{-6}$ s,

$$1 - \frac{v_\mu^{min}}{c} \simeq \frac{(2.2 \times 10^{-6})^2}{2(879.4)^2}$$

$$1 - \frac{v_\mu^{min}}{c} \simeq 3.13 \times 10^{-18}$$